Machine Learning for DNA Artifact Classification
Low Template DNA (LTDNA) Analysis

EXECUTIVE SUMMARY REPORT

Dataset: Mix Profiles
Date: December 17, 2025
Models Evaluated: 11 (9 individual + 2 ensembles)

KEY FINDINGS

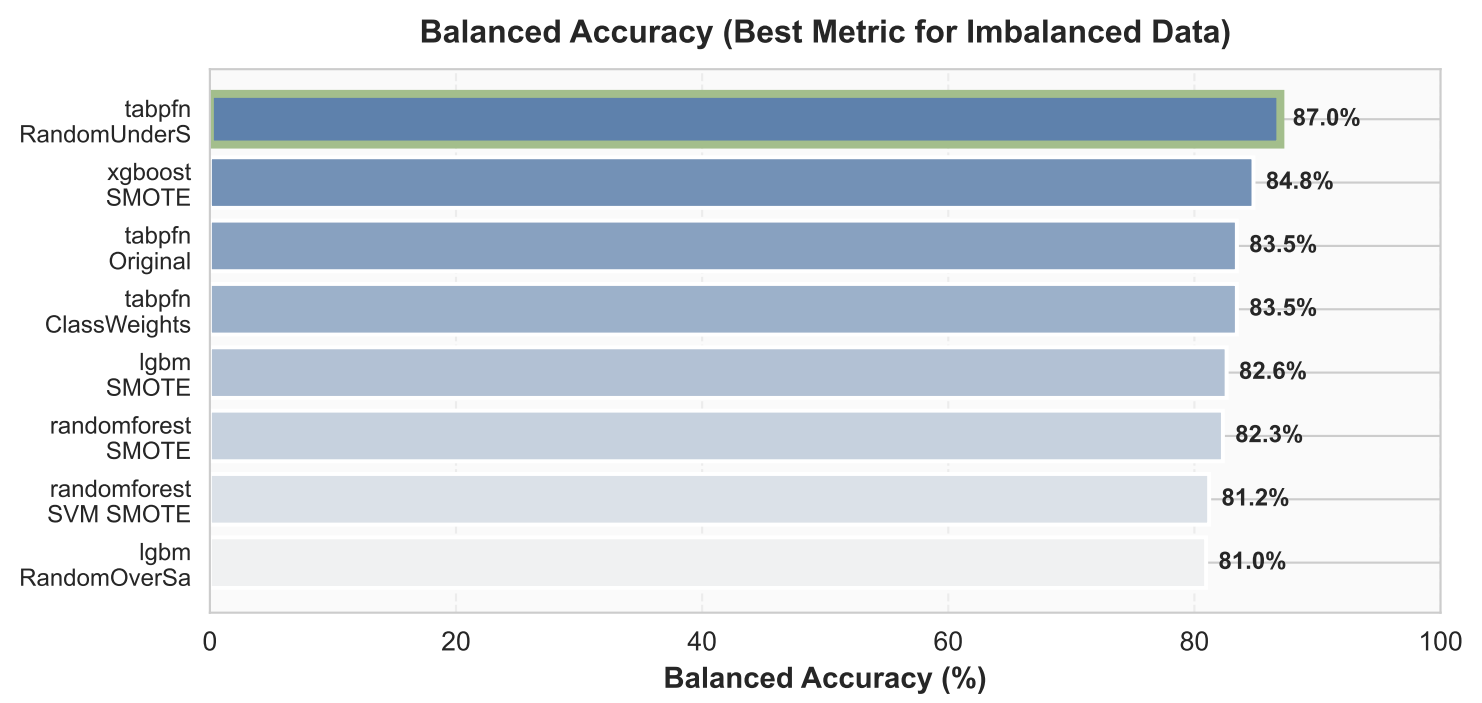
Best Model: tabpfn with RandomUnderSampler

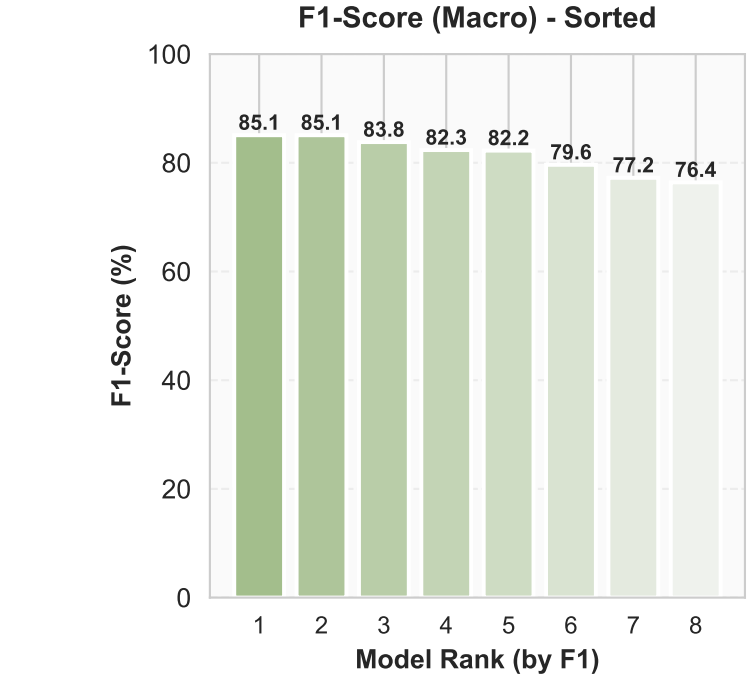
Performance Metrics:
- Balanced Accuracy: 86.98%
- F1-Score (macro): 82.25%
- Overall Accuracy: 82.56%

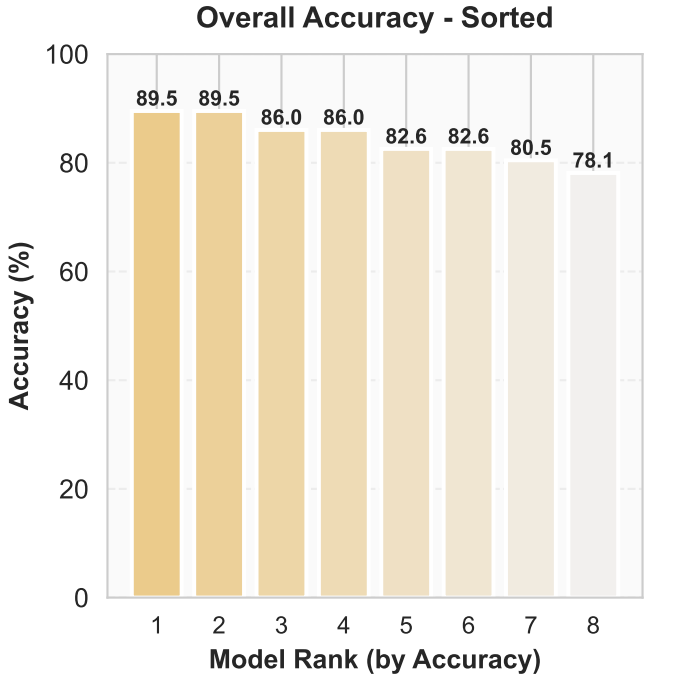
Classification Task:
- 4 Classes: Allele, Back_Stutter, Forward_Stutter, Pull_Up
- 426 training samples with class imbalance
- 307 original features + 56 engineered features

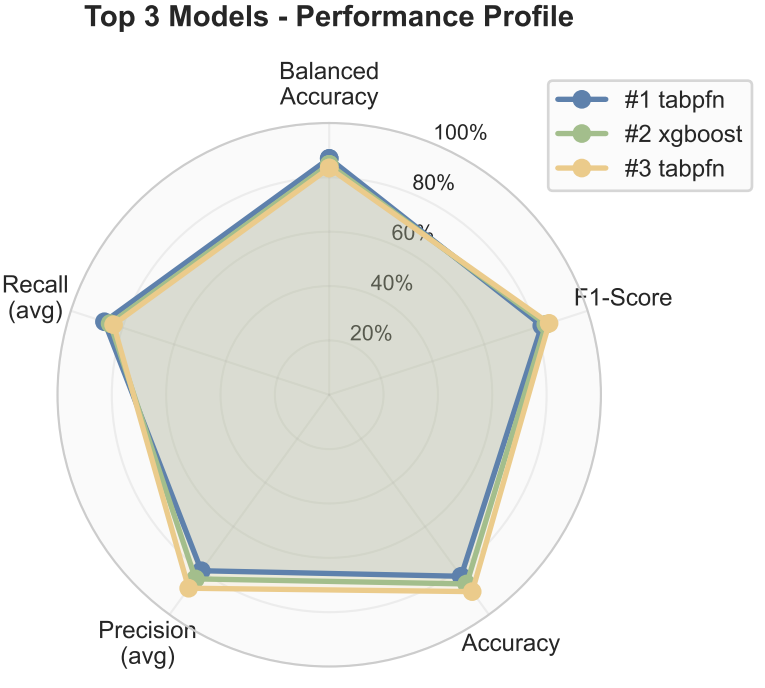
Optimization Approach:
✓ Feature scaling (StandardScaler)
✓ Class weight balancing
✓ Early stopping for gradient boosting
✓ Ensemble methods (Stacking & Voting)
✓ Multiple resampling strategies tested

This report presents the essential results and visualizations from comprehensive machine learning analysis for LTDNA artifact classification.

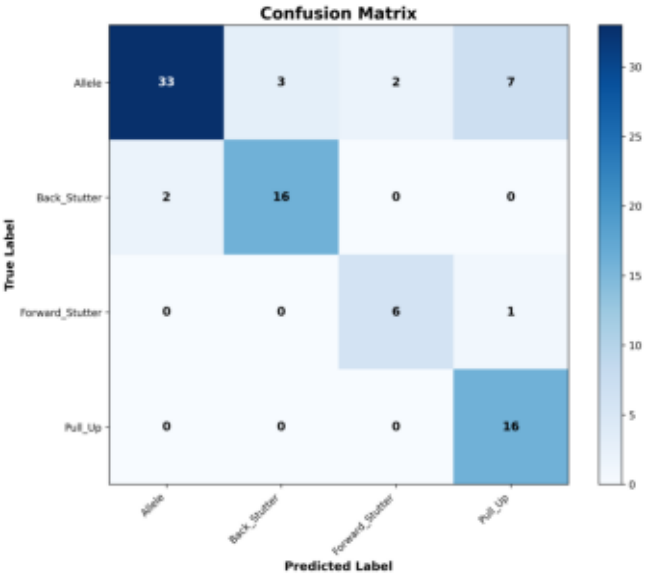

Balanced Accuracy (Best Metric for Imbalanced Data)
tabpfn RandomUnderS
xgboost SMOTE
tabpfn Original
tabpfn ClassWeights
lgbm SMOTE
randomforest SMOTE
randomforest SVM SMOTE
lgbm RandomOverSa
87.0%
84.8%
83.5%
83.5%
82.6%
82.3%
81.2%
81.0%
0
20
40
60
80
100
Balanced Accuracy (%)


F1-Score (Macro) - Sorted
F1-Score (%)
0
20
40
60
80
100
85.1
85.1
83.8
82.3
82.2
79.6
77.2
76.4
1
2
3
4
5
6
7
8
Model Rank (by F1)

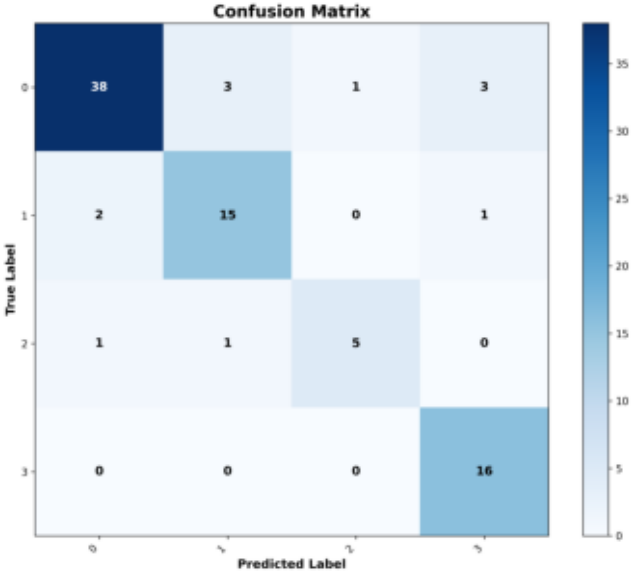

Overall Accuracy - Sorted
Accuracy (%)
0
20
40
60
80
100
89.5
89.5
86.0
86.0
82.6
82.6
80.5
78.1
1
2
3
4
5
6
7
8
Model Rank (by Accuracy)


Top 3 Models - Performance Profile
Balanced Accuracy
100%
80%
60%
40%
20%
F1-Score
Accuracy
Precision (avg)
Recall (avg)
#1 tabpfn
#2 xgboost
#3 tabpfn

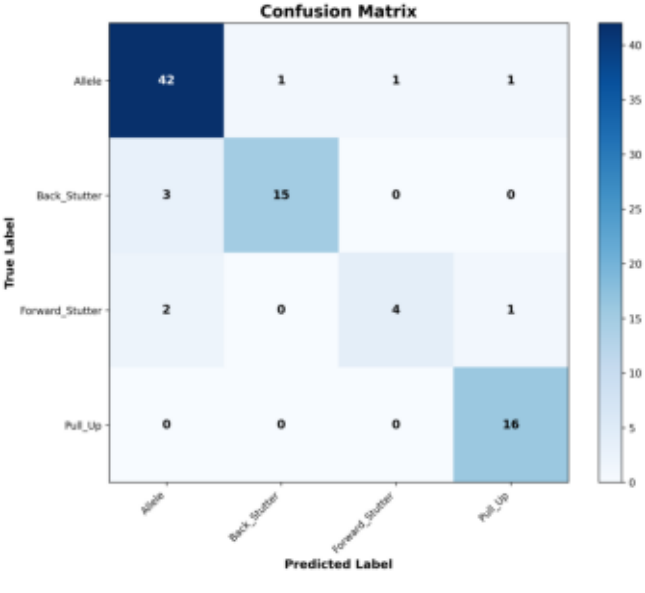
#1: TABPFN with RandomUnderSampler
Balanced Acc: 86.98% | F1: 82.25%



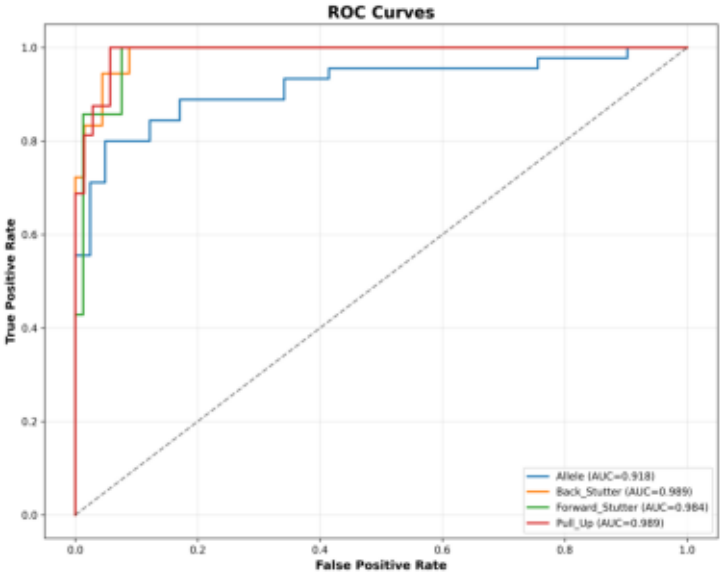
#2: XGBOOST with SMOTE
Balanced Acc: 84.80% | F1: 83.82%



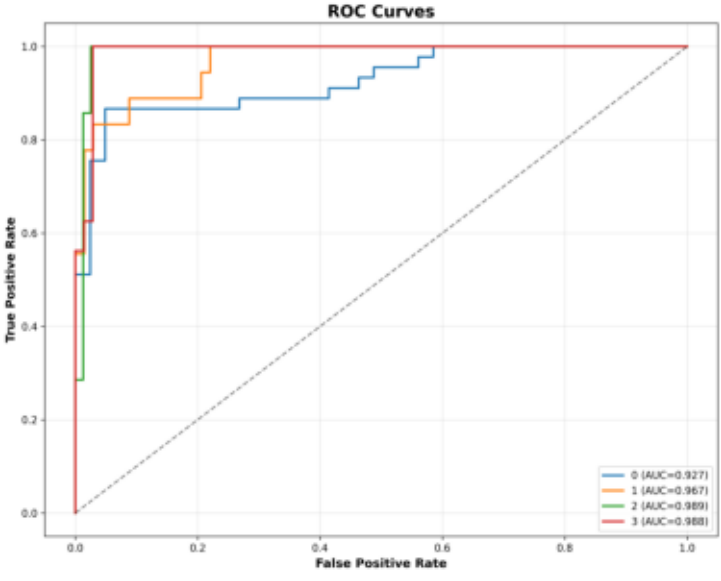
#3: TABPFN with Original
Balanced Acc: 83.45% | F1: 85.08%



#1: TABPFN with RandomUnderSampler



#2: XGBOOST with SMOTE



#3: TABPFN with Original

